

Sebin Jung

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RESEARCH INTERESTS

My research interests lie at the intersection of **robot learning, optimal control, and safety**. I aim to develop **principled, control theory-informed learning methods** for adaptive, reliable, and high-performance robotic systems, bridging theoretical advances with validation on **whole-body scale hardware**.

EDUCATION

Carnegie Mellon University

Master of Science in Mechanical Engineering - Research | GPA: 4.00/4.00

May, 2026

Pittsburgh, PA

- Advisors: Dr. Changliu Liu (Robotics Institute), Dr. John M. Dolan (Robotics Institute)

Kyungpook National University

Bachelor of Science in Mechanical Engineering

Feb. 2024

Daegu, South Korea

University of California, Berkeley

Summer Session

Jul. - Aug. 2022

Berkeley, CA

PUBLICATIONS & PREPRINTS

C=CONFERENCE PAPERS AND SUBMISSIONS, P=PREPRINTS

* Equal Contribution

- [C.1.] **Sebin Jung**, Abulikemu Abuduweili, Jiaxing Li, Changliu Liu, **Whole-Body Safe Control of Robotic Systems with Koopman Neural Dynamics**. In *Proceedings of the 2026 IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, 2026. **Oral Presentation**.
- [C.2.] **Sebin Jung***, Maitham F. AL-Sunni*, Juan Alvarez-Padilla, Zachary Manchester, Changliu Liu, John M. Dolan, **LLA-MPPI: Rapidly Adaptive Whole-body Control of Legged Robots with GPU-Accelerated Parallel Simulations**. Under review at the *2027 IEEE International Conference on Robotics and Automation (ICRA)*, 2027.
- [C.3.] Seokhyun Kim, Jin Woo Hwang, Itak Choi, Chan Kyu Lee, Kunwoo Lee, **Sebin Jung**, Avery Williamson, Victoria Webster-Wood, **Meow-Mentum: A Multi-Strategy Aerial Self-Righting Robot for Evaluating the Contribution of an Actuated Tail Across Fall Conditions**. Under review at the *2027 IEEE International Conference on Robotics and Automation (ICRA)*, 2027.
- [C.4.] Yifan Sun, Rui Chen, Kai S. Yun, Yikuan Fang, **Sebin Jung**, Weiye Zhao, Changliu Liu, **Safe Humanoid Autonomy and Teleoperation with Safe Set Algorithm**. Extended abstract accepted to *Joint 10th IFAC Symposium on Mechatronic Systems and 14th Symposium on Robotics*, 2025.
- [P.1.] Yifan Sun, Rui Chen, Kai S. Yun, Yikuan Fang, **Sebin Jung**, Feihan Li, Bowei Li, Weiye Zhao, Changliu Liu, **SPARK: A Modular Benchmark for Humanoid Robot Safety**. Preprint, 2025. **Video**.

HONORS & SCHOLARSHIPS

Best Master of Science Research Paper Award, Carnegie Mellon University

May, 2026

Outstanding Learner, Hyundai Motor Group,

Feb. 2024

Excellent achievement award for *H-Mobility Class*

1st-Place Award in Graduation Project, Kyungpook National University,

Spring 2023

Best research award for *Capstone Design of Mechanical Engineering Project 1 course*

Scholarship Recommended by Dean, Kyungpook National University

2022, Spring 2023

Scholarship of Academic Excellence, Kyungpook National University

Fall 2021 - Spring 2023

RESEARCH EXPERIENCE

Intelligent Control Lab (CMU Robotics Institute)

Oct. 2024 – Present

Advisor: [Dr. Changliu Liu](#) | Roles: Research Assistant, Lab Manager

Pittsburgh, PA

- Led development of real-time whole-body safe controller for robotic arms and legged robots using learned Koopman dynamics representation [C.1.]
 - Built OSQP-based unified quadratic program framework that combines task objective with safety constraints
 - Developed PyTorch-based adversarial training pipeline to improve the robustness of learned safety constraints
 - Validated controller on Kinova Gen3 arm and Unitree Go2, achieving over $4.2\times$ faster computation while improving safety-performance trade-off
- Co-developed SPARK, benchmark for testing robot safety using standardized metrics, safe controllers, and simulated system failures [C.4., P.1.]
 - Built MuJoCo simulation and hardware evaluation pipelines for second-order dynamics representation of Unitree G1 humanoid robot
 - Integrated Kinova Gen3 robotic arm into the evaluation pipeline, expanding benchmark beyond humanoids

- Led development of LLA-MPPI, real-time whole-body adaptive controller that enables zero-shot rapid adaptation towards abrupt environment changes [C.2.]
 - Built GPU-accelerated dynamics ensembling method with MuJoCo Warp to identify changing robot and environment dynamics online
 - Developed multi-threaded sampling-based controller to optimize whole-body robot motion in real time
 - Validated the method on Unitree Go2 hardware, and achieved 97.5% success rate across four simulated tasks, compared with 74% for the strongest baseline

- Developed multi-policy, multi-objective reinforcement learning framework for autonomous vehicle drifting
 - Trained twin Soft Actor-Critic (SAC) policies for lane following and drift control
 - Designed a high-level neural decision module to switch between policies based on road curvature
 - Validated performance in CARLA simulator
- Developed model-free reinforcement learning controller for autonomous driving
 - Trained continuous-control policy using Twin Delayed Deep Deterministic Policy Gradient (TD3)
 - Formulated Stanley-inspired reward function penalizing cross-track and heading-angle errors
 - Validated learned policy in IPG CarMaker, industry-level vehicle dynamics simulator

LEADERSHIP EXPERIENCE

- Sergeant discharged
- Completed Sikorsky UH-60 maintenance engineering and crew training courses
- Participated in maintenance operations of UH-60, KUH-1, wartime condition helicopter maintenance drill

SKILLS

- **Language/Tools:** Python, C/C++, Julia, MATLAB, Simulink, Bash, Git, ROS1/2, Docker, CUDA, Creo, $\LaTeX$
- **Libraries:** PyTorch, Brax, JAX, MJX, Warp, MJWarp, OSQP, scikit-learn, NumPy, SciPy, matplotlib, Gazebo, MuJoCo, MuJoCo Playground, IsaacSim, PyBullet, CARLA, Webots, TensorBoard, Wandb
- **Robots/Hardware:** Unitree Go2, Unitree G1, Kinova, F1Tenth, Jetson, OptiTrack, Bambu Lab H2S
- **Languages:** Fully professional in **English**(TOEFL iBT 112), Native in **Korean**

SELECTED COURSES

Carnegie Mellon University

Intermediate Deep Learning (10-617), Nonlinear Control (24-776), Provably Safe Robotics (16-883), Optimal Control and Reinforcement Learning (16-745), Modern Control Theory (24-677), Machine Learning and Artificial Intelligence for Engineers (24-787)

Kyungpook National University

Optimization Theory (ELEC 0842), System Control (MECH 0482), Mechatronics (MECH 0358), Kinematics & Mechanics of Machinery (MECH 0356), Numerical Analysis (STAT 0452), Dynamics (MECH 0225)

ACADEMIC PROJECTS

Modern Control and Estimation for Autonomous Robots

2024, CMU

- Designed advanced control systems for autonomous vehicles and aerial robots using state-space representations
- Implemented full-state feedback and LQR controllers for trajectory tracking and stability under disturbances
- Developed Extended Kalman Filter (EKF) SLAM for localization and mapping with uncertain sensor data
- Applied Model Reference Adaptive Control (MRAC) to enable quadrotor flight under partial actuator failure

Quanser Self-Driving Car Student Competition (Won 4th Place)

2024, ACC

- Student challenge for designing autonomous driving pipeline in urban traffic
- Combined Pure Pursuit and adaptive speed control to minimize lap time
- Integrated YOLOv7 camera perception and LiDAR scan for obstacle and traffic detection

F1Tenth Korea Championship

2023, ICCAS

- Performed SLAM using the Iterative Closest Point (ICP) algorithm
- Optimized global trajectory to minimize lap time and exploit terrain information
- Applied Pure Pursuit control with a finely tuned speed profile for high-speed racing